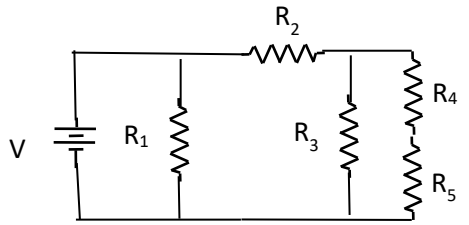
Physics 2 Problem Solving – Chapter 43 – Series/parallel combinations.

1) Use Ohms Law and series/parallel combinations to solve.



V = 50 V
R_1 = 20 Ω
R_2 = 10 Ω
R_3 = 30 Ω
R_4 = 10 Ω
R_5 = 20 Ω

A) Determine the equivalent resistance of the circuit and the current supplied to the circuit by the power supply.

Answer: 11.1 Ω and 4.5 A

B) Determine the current flowing through resistor 1.

Answer: 2.5 A

C) Determine the current flowing through resistor 5 and the voltage drop across resistor 5.

Answer: 1 A and 20 V